GRADE 7 ACCELERATED · UNIT 4

Solving Problems with Rational Numbers

Six lessons in teaching order · the next slide jumps to each one

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GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

MA.8.NSO.1.5

I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

Warm-Up

1. Write 0.00062 in scientific notation.
2. How many times larger is 8×10^7 than 2×10^4 ?
3. Find the value: $\sqrt{144}$
4. Find the sum: $0.49 + 1.2$

Warm-Up

1. Write 0.00062 in scientific notation. **6.2×10^{-4}**
2. How many times larger is 8×10^7 than 2×10^4 ? **4,000**
3. Find the value: $\sqrt{144}$ **12**
4. Find the sum: $0.49 + 1.2$ **1.69**

Vocabulary

scientific notation

a number written as $a \times 10^n$,

where a is at least 1 and less than 10 and n is an integer

coefficient

the factor a in $a \times 10^n$. In a finished answer it always satisfies $1 \le a < 10$

common power of ten

the matching power that two numbers are written over before they are added or subtracted — the place value they share

Notes

I. A power of ten is a place value

10^3	10^4	10^5	10^6
thousands	ten thousands	hundred thousands	millions

$$49{,}500 + 121{,}000 = 170{,}500$$

$$4.95 \times 10^4 + 12.1 \times 10^4 = 17.05 \times 10^4 = 1.705 \times 10^5$$

Two numbers can be added only when they carry the same power of ten.

Notes

II. Same power: add or subtract the coefficients

$$(5.2 \times 10^{6}) + (3.1 \times 10^{6}) = 8.3 \times 10^{6}$$

$$(7 \times 10^{-8}) + (6 \times 10^{-8}) = 13 \times 10^{-8} = 1.3 \times 10^{-7}$$

Add or subtract the coefficients. The power of ten does not change.

Then check the coefficient: if it is not between 1 and 10, the answer is not finished.

Notes

III. Different powers: rewrite one of them first

$$(9.4 \times 10^5) - (8.2 \times 10^3)$$

$$= (9.4 \times 10^5) - (0.082 \times 10^5) = 9.318 \times 10^5$$

$$(1.3 \times 10^3) + (3.4 \times 10^5) = 3.413 \times 10^5, \quad \text{never} \quad 4.7 \times 10^8$$

Rewrite one number so both carry the same power of ten, then add or subtract the coefficients.

Exponent up one, decimal point one place left. The value does not move.

Example 1

$$(8.6 \times 10^7) + (5.9 \times 10^7)$$

Add. Write the answer in scientific notation.

Example 1

$$(8.6 \times 10^{7}) + (5.9 \times 10^{7}) = 14.5 \times 10^{7}$$

$$14.5 \times 10^{7} = 1.45 \times 10^{8}$$

$$145{,}000{,}000$$

Answer: 1.45×10^{8}

Your Turn

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

Add. Write the answer in scientific notation.

Your Turn

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

Add. Write the answer in scientific notation.

Same power, so add the fronts: $4.8 + 7.5 = 12.3$

$12.3 \times 10^{-3} = 1.23 \times 10^{-2}$

Answer: 1.23×10^{-2}

Example 2

$$(7.3 \times 10^6) - (4.5 \times 10^5)$$

Subtract. Write the answer in scientific notation.

Example 2

$4.5 \times 10^5 = 0.45 \times 10^6$

$(7.3 \times 10^6) - (0.45 \times 10^6) = 6.85 \times 10^6$

$7{,}300{,}000 - 450{,}000 = 6{,}850{,}000$

Answer: 6.85×10^6

Your Turn

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

Subtract. Write the answer in scientific notation.

Your Turn

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

Subtract. Write the answer in scientific notation.

Match the powers: $9.0 \times 10^{-6} = 0.09 \times 10^{-4}$

$5.6 - 0.09 = 5.51$

5.51×10^{-4}

Answer: 5.51×10^{-4}

Whiteboards · Question 1 of 9

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

Answer it.

Whiteboards · Question 1 of 9

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

Same power, so add the fronts: $3.2 + 4.1 = 7.3$

7.3×10^5

Answer: 7.3×10^5

Whiteboards · Question 2 of 9

A stadium sold 4.2×10^4 tickets on Friday

and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

Answer it.

Whiteboards · Question 2 of 9

A stadium sold 4.2×10^4 tickets on Friday

and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

Same power, so add the fronts: $4.2 + 3.5 = 7.7$

7.7×10^4

Answer: 7.7×10^4

Whiteboards · Question 3 of 9

$$(8 \times 10^{-6}) + (3 \times 10^{-6})$$

Answer it.

Whiteboards · Question 3 of 9

$$(8 \times 10^{-6}) + (3 \times 10^{-6})$$

$$8 + 3 = 11$$

$$11 \times 10^{-6} = 1.1 \times 10^{-5}$$

Answer: 1.1×10^{-5}

Whiteboards · Question 4 of 9

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

Answer it.

Whiteboards · Question 4 of 9

$$(5.6 \times 10^{8}) - (4.9 \times 10^{8})$$

$$5.6 - 4.9 = 0.7$$

$$0.7 \times 10^{8} = 7 \times 10^{7}$$

Answer: 7×10^{7}

Whiteboards · Question 5 of 9

$$(1.6 \times 10^4) + (2.3 \times 10^6)$$

Which is the sum?

A. 3.9×10^6

B. 3.9×10^{10}

C. 2.316×10^6

D. 2.46×10^6

Answer it.

Whiteboards · Question 5 of 9

$$(1.6 \times 10^4) + (2.3 \times 10^6)$$

Which is the sum?

Match the powers: $1.6 \times 10^4 = 0.016 \times 10^6$

$0.016 + 2.3 = 2.316$

2.316×10^6

Answer: C — 2.316×10^6

Whiteboards · Question 6 of 9

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

A. 12.3×10^6

B. 12.3×10^{12}

C. 1.23×10^6

D. 1.23×10^7

Answer it.

Whiteboards · Question 6 of 9

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

$$2.5 + 9.8 = 12.3$$

$$12.3 \times 10^6 = 1.23 \times 10^7$$

Answer: D — 1.23×10^7

Whiteboards · Question 7 of 9

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

Answer it.

Whiteboards · Question 7 of 9

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

Match the powers: $6.4 \times 10^{-3} = 0.64 \times 10^{-2}$

$7.8 - 0.64 = 7.16$

7.16×10^{-2}

Answer: 7.16×10^{-2}

Whiteboards · Question 8 of 9

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

Answer it.

Whiteboards · Question 8 of 9

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

$$1.02 \times 10^7 = 10.2 \times 10^6$$

$$4.5 + n = 10.2$$

$$n = 10.2 - 4.5 = 5.7$$

Answer: $n = 5.7$

Whiteboards · Question 9 of 9

Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$.

Name the two mistakes he made, and give the correct sum.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$.

Name the two mistakes he made, and give the correct sum.

The powers do not match, and exponents are not added.

$1.9 \times 10^2 = 0.019 \times 10^4$

$0.019 + 6.6 = 6.619$, so 6.619×10^4

Answer: 6.619×10^4

Independent Set

1. $(4.1 \times 10^{6}) + (2.7 \times 10^{6})$
2. $(9.3 \times 10^{-5}) - (5.1 \times 10^{-5})$
3. $(7.5 \times 10^{3}) + (8.9 \times 10^{3})$
4. $(6.2 \times 10^{7}) + (3.5 \times 10^{6})$
5. $(4.8 \times 10^{-2}) - (9.0 \times 10^{-4})$
6. Find n. $(5.2 \times 10^{9}) + (n \times 10^{9}) = 1.31 \times 10^{10}$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due by the Unit 4 test.

Today's skills — all required

Add and subtract numbers written in scientific notation — HUR

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

MA.8.NSO.1.5

I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

Warm-Up

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$
2. Write as one power: $\frac{10^9}{10^4}$
3. Find the value: $\sqrt{169}$
4. Find the quotient: $\frac{4.8}{1.6}$

Warm-Up

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$ **6.2×10^6**
2. Write as one power: $\frac{10^9}{10^4}$ **10^5**
3. Find the value: $\sqrt{169}$ **13**
4. Find the quotient: $\frac{4.8}{1.6}$ **3**

Vocabulary

product of powers law

$10^m \cdot 10^n = 10^{m+n}$ — the law from 3.01,

now doing the work on the powers of ten

quotient of powers law

$\frac{10^m}{10^n} = 10^{m-n}$ — top minus bottom, from 3.02

regrouping a product

using the commutative and associative properties to write $(2 \times 10^4)(3 \times 10^5)$

as $(2 \cdot 3)(10^4 \cdot 10^5)$

Notes

I. Multiplying: regroup, then use the laws

$$(2 \times 10^4)(3 \times 10^5) = (2 \cdot 3)(10^4 \cdot 10^5) = 6 \times 10^9$$

$$6 \times 10^9, \quad \text{never} \quad 6 \times 10^{20}$$

Multiply the coefficients. Add the exponents — that is the product of powers law, not a new rule.

Notes

II. Dividing: the same move with the quotient law

$$\frac{8.4 \times 10^{9}}{2.1 \times 10^{4}} = \frac{8.4}{2.1} \times 10^{9-4} = 4 \times 10^{5}$$

$$\frac{3 \times 10^{8}}{3 \times 10^{-4}} = 1 \times 10^{8-(-4)} = 1 \times 10^{12}$$

Divide the coefficients. Subtract the exponents, top minus bottom.

Notes

III. Finish the answer — both directions

$$(8 \times 10^{2})(7.5 \times 10^{4}) = 60 \times 10^{6} = 6 \times 10^{7}$$

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}} = 0.53 \times 10^{-2} = 5.3 \times 10^{-3}$$

Every answer ends with a coefficient at least 1 and less than 10.

Example 1

$$(4.5 \times 10^{-3})(6 \times 10^{7})$$

Multiply. Write the answer in scientific notation.

Example 1

$$(4.5 \times 10^{-3})(6 \times 10^{7}) = (4.5 \cdot 6)(10^{-3} \cdot 10^{7})$$

$$= 27 \times 10^{4}$$

$$27 \times 10^{4} = 2.7 \times 10^{5}$$

Answer: 2.7×10^{5}

Your Turn

$$(2.5 \times 10^{6})(8 \times 10^{-2})$$

Multiply. Write the answer in scientific notation.

Your Turn

$$(2.5 \times 10^{6})(8 \times 10^{-2})$$

Multiply. Write the answer in scientific notation.

Multiply the fronts: $2.5 \times 8 = 20$

Add the exponents: $10^{6} \times 10^{-2} = 10^{4}$

$20 \times 10^{4} = 2 \times 10^{5}$

Answer: 2×10^{5}

Example 2

$$\frac{(8 \times 10^2)(7.5 \times 10^4)}{5 \times 10^2}$$

Evaluate. Write the answer in scientific notation.

Example 2

$$(8 \times 10^{2})(7.5 \times 10^{4}) = 60 \times 10^{6}$$

$$\frac{60 \times 10^{6}}{5 \times 10^{2}} = 12 \times 10^{4}$$

$$12 \times 10^{4} = 1.2 \times 10^{5}$$

Answer: 1.2×10^{5}

Your Turn

$$\frac{(6 \times 10^{5})(4 \times 10^{-3})}{8 \times 10^{4}}$$

Evaluate. Write the answer in scientific notation.

Your Turn

$$\frac{(6 \times 10^{5})(4 \times 10^{-3})}{8 \times 10^{4}}$$

Evaluate. Write the answer in scientific notation.

Top: $6 \times 4 = 24$ and $10^{5} \times 10^{-3} = 10^{2}$

Divide: $24 \div 8 = 3$ and $10^{2} \div 10^{4} = 10^{-2}$

3×10^{-2}

Answer: 3×10^{-2}

Whiteboards · Question 1 of 9

$$(3 \times 10^5)(3 \times 10^6)$$

Answer it.

Whiteboards · Question 1 of 9

$$(3 \times 10^5)(3 \times 10^6)$$

Multiply the fronts: $3 \times 3 = 9$

Add the exponents: $10^5 \times 10^6 = 10^{11}$

9×10^{11}

Answer: 9×10^{11}

Whiteboards · Question 2 of 9

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

Answer it.

Whiteboards · Question 2 of 9

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

Multiply the fronts: $4 \times 2.5 = 10$

Add the exponents: $10^3 \times 10^2 = 10^5$

$10 \times 10^5 = 1 \times 10^6$

Answer: 1×10^6

Whiteboards · Question 3 of 9

$$\frac{9.6 \times 10^{8}}{1.2 \times 10^{3}}$$

Answer it.

Whiteboards · Question 3 of 9

$$\frac{9.6 \times 10^{8}}{1.2 \times 10^{3}}$$

Divide the fronts: $9.6 \div 1.2 = 8$

Subtract the exponents: $10^{8} \div 10^{3} = 10^{5}$

8×10^{5}

Answer: 8×10^{5}

Whiteboards · Question 4 of 9

$$\frac{3.28 \times 10^{7}}{4 \times 10^{10}}$$

Answer it.

Whiteboards · Question 4 of 9

$$\frac{3.28 \times 10^{7}}{4 \times 10^{10}}$$

Divide the fronts: $3.28 \div 4 = 0.82$

Subtract the exponents: $10^{7} \div 10^{10} = 10^{-3}$

$0.82 \times 10^{-3} = 8.2 \times 10^{-4}$

Answer: 8.2×10^{-4}

Whiteboards · Question 5 of 9

$$(4 \times 10^{7})(2 \times 10^{9})$$

Which is the product?

A. 8×10^{-2}

B. 8×10^{63}

C. 8×10^{16}

D. 6×10^{16}

Answer it.

Whiteboards · Question 5 of 9

$$(4 \times 10^{7})(2 \times 10^{9})$$

Which is the product?

Multiply the fronts: $4 \times 2 = 8$

Add the exponents: $10^{7} \times 10^{9} = 10^{16}$

8×10^{16}

Answer: C — 8×10^{16}

Whiteboards · Question 6 of 9

$$(5 \times 10^6)(4 \times 10^3)$$

Which is the product, written in scientific notation?

A. 2×10^9

B. 20×10^9

C. 20×10^{18}

D. 2×10^{10}

Answer it.

Whiteboards · Question 6 of 9

$$(5 \times 10^{6})(4 \times 10^{3})$$

Which is the product, written in scientific notation?

Multiply the fronts: $5 \times 4 = 20$

Add the exponents: $10^{6} \times 10^{3} = 10^{9}$

$20 \times 10^{9} = 2 \times 10^{10}$

Answer: D — 2×10^{10}

Whiteboards · Question 7 of 9

$$\frac{7 \times 10^{9}}{7 \times 10^{-6}}$$

Answer it.

Whiteboards · Question 7 of 9

$$\frac{7 \times 10^{9}}{7 \times 10^{-6}}$$

Divide the fronts: $7 \div 7 = 1$

Subtract the exponents: $10^{9} \div 10^{-6} = 10^{9-(-6)} = 10^{15}$

1×10^{15}

Answer: 1×10^{15}

Whiteboards · Question 8 of 9

$$\frac{8.4 \times 10^{7}}{n \times 10^{2}} = 4.2 \times 10^{5}$$

What is the value of n ?

Answer it.

Whiteboards · Question 8 of 9

$$\frac{8.4 \times 10^{7}}{n \times 10^{2}} = 4.2 \times 10^{5}$$

What is the value of n ?

Fronts: $8.4 \div n = 4.2$, so $n = 8.4 \div 4.2 = 2$

Check: $\frac{8.4 \times 10^{7}}{2 \times 10^{2}} = 4.2 \times 10^{5}$

Answer: $n = 2$

Whiteboards · Question 9 of 9

For $\frac{7.2 \times 10^{11}}{2.4 \times 10^{3}}$ Marcus writes 4.8×10^{8}

and Sara writes 3×10^{14}

Name what each one did wrong, and give the correct quotient.

Write your answer in sentences.

Whiteboards · Question 9 of 9

For $\frac{7.2 \times 10^{11}}{2.4 \times 10^{3}}$ Marcus writes 4.8×10^{8}

and Sara writes 3×10^{14}

Name what each one did wrong, and give the correct quotient.

Marcus subtracted the fronts. Sara added the exponents.

$7.2 \div 2.4 = 3$ and $10^{11} \div 10^{3} = 10^{8}$

3×10^{8}

Answer: 3×10^{8}

Independent Set

1. $(2 \times 10^5)(4 \times 10^3)$
2. $\dfrac{9.9 \times 10^7}{3.3 \times 10^2}$
3. $(5 \times 10^6)(6 \times 10^{-2})$
4. $\dfrac{1.8 \times 10^3}{6 \times 10^7}$
5. $\dfrac{4 \times 10^9}{8 \times 10^{-2}}$
6. Find n. $(2 \times 10^n)(3 \times 10^4) = 6 \times 10^{10}$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due by the Unit 4 test.

Today's skills — all required

Multiply numbers written in scientific notation — YZU
Divide numbers written in scientific notation — SGT

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

MA.8.NSO.1.6

I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

Warm-Up

1. $(4 \times 10^5)(3 \times 10^{-2})$
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$
3. Write 0.00902 in scientific notation.
4. Round 4.738 to the nearest hundredth.

Warm-Up

1. $(4 \times 10^5)(3 \times 10^{-2})$ **1.2×10^4**
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$ **5.89×10^7**
3. Write 0.00902 in scientific notation. **9.02×10^{-3}**
4. Round 4.738 to the nearest hundredth. **4.74**

Vocabulary

significant digits

the digits of a measurement that carry information: every nonzero digit, every zero between nonzero digits, and a trailing zero only when a decimal point is written

leading zero

a zero before the first nonzero digit. 0.0047 has two leading zeros and two significant digits

measurement

a quantity that was measured, so its last digit is uncertain. A count is not a measurement — 24 students is exactly 24, and 60 seconds in a minute is exact

Notes

I. Which digits are significant

Measurement	Digits	Why
4,207	4	the middle zero counts
0.0058	2	leading zeros never count
3.200	4	trailing zeros count after a decimal point
52,000	2	no decimal point, so they do not

Notes

II. Multiplying or dividing: count the digits

$$(2.5 \times 10^{2})(4.13 \times 10^{2}) = 1.0325 \times 10^{5}$$

2 significant digits and 3, so the answer keeps 2: 1.0×10^{5}

Multiply or divide: the answer takes the smaller of the two significant-digit counts.

Notes

III. Adding or subtracting: a different rule

$$1{,}234 + 5.6 = 1{,}239.6$$

1,234 stops at the ones place; 5.6 stops at the tenths

so the sum stops at the ones place: $1{,}240 = 1.240 \times 10^3$

Add or subtract: the answer stops at the leftmost of the two last-digit places.

Example 1

A solar array measures 1.25×10^{2} m by 8.6×10^{1} m.

What is its area?

Give the answer to the correct number of significant digits.

Example 1

$(1.25 \times 10^{2})(8.6 \times 10^{1}) = 10{,}750$

1.25 has 3 digits, 8.6 has 2 digits

$10{,}750 \approx 11{,}000 = 1.1 \times 10^{4}$

Answer: 1.1×10^{4} square meters

Your Turn

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

Your Turn

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

$4.6 \times 2.35 = 10.81$ and $10^2 \times 10^2 = 10^4$

$10.81 \times 10^4 = 108{,}100$

Keep 2 significant digits (4.6 has 2): 1.1×10^5

Answer: 1.1×10^5 square meters

Example 2

A container holds 3.09×10^{-3} cubic feet of liquid.

3.472×10^{-5} cubic feet evaporates.

How much is left? Give the answer with the correct precision.

Example 2

$0.00309 - 0.00003472 = 0.00305528$

0.00309 stops at 10^{-5}

$0.00305528 \approx 0.00306 = 3.06 \times 10^{-3}$

Answer: 3.06×10^{-3} cubic feet

Your Turn

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

Your Turn

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

$3{,}850 - 274 = 3{,}576$

3,850 stops at the tens place, so round to tens: 3,580

$3{,}580 = 3.58 \times 10^3$

Answer: 3.58×10^3 m

Whiteboards · Question 1 of 9

How many significant digits are in 0.00470 ?

Answer it.

Whiteboards · Question 1 of 9

How many significant digits are in 0.00470 ?

Leading zeros do not count: 0.00

4, 7 and the final 0 count: 3 significant digits

Answer: 3

Whiteboards · Question 2 of 9

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 meters deep.

Find the surface area. Round to the correct number of significant digits.

Answer it.

Whiteboards · Question 2 of 9

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 meters deep.

Find the surface area. Round to the correct number of significant digits.

$1.4 \times 8.5 = 11.9$ and $10^2 \times 10^1 = 10^3$

$11.9 \times 10^3 = 11{,}900$

Keep 2 significant digits: 1.2×10^4

Answer: 1.2×10^4 square meters

Whiteboards · Question 3 of 9

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5.

How should the answer be written?

Answer it.

Whiteboards · Question 3 of 9

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5.

How should the answer be written?

The answer keeps the smaller count: 3 significant digits.

7.4126×10^5 becomes 7.41×10^5

Answer: 7.41×10^5

Whiteboards · Question 4 of 9

$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured.

Write the sum with the correct precision.

Answer it.

Whiteboards · Question 4 of 9

$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured.

Write the sum with the correct precision.

2,500 + 471 = 2,971

2,500 stops at the hundreds place, so round to hundreds: 3,000

$3{,}000 = 3.0 \times 10^3$

Answer: 3.0×10^3 m

Whiteboards · Question 5 of 9

How many significant digits are in 0.03080 ?

A. 4

B. 2

C. 5

D. 3

Answer it.

Whiteboards · Question 5 of 9

How many significant digits are in 0.03080 ?

Leading zeros do not count: 0.0

3, 0, 8 and the final 0 count: 4 significant digits

Answer: A — 4

Whiteboards · Question 6 of 9

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

A. 26.1×10^4 cm^2

B. 2.6×10^5 cm^2

C. 2.6×10^4 cm^2

D. 2.61×10^5 cm^2

Answer it.

Whiteboards · Question 6 of 9

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

$6.0 \times 4.35 = 26.1$ and $10^2 \times 10^2 = 10^4$

$26.1 \times 10^4 = 261{,}000$

Keep 2 significant digits (6.0 has 2): 2.6×10^5

Answer: B — 2.6×10^5 cm²

Whiteboards · Question 7 of 9

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

Answer it.

Whiteboards · Question 7 of 9

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

$420 \div 3.0 = 140$

3.0 has 2 significant digits, so keep 2: 1.4×10^2

Answer: 1.4×10^2 km/h

Whiteboards · Question 8 of 9

A measured length times a measured width gives an area of 3.6×10^3 m^2. The length measured 9.0×10^1 m.

What did the width measure, to two significant digits?

Answer it.

Whiteboards · Question 8 of 9

A measured length times a measured width gives an area of 3.6×10^3 m². The length measured 9.0×10^1 m.

What did the width measure, to two significant digits?

$3{,}600 \div 90 = 40$

Two significant digits: 4.0×10^1

Answer: 4.0×10^1 m

Whiteboards · Question 9 of 9

Ellis says: 9,145 has 4 significant digits and 4.8 has 2,

so the sum $9{,}145 + 4.8$ must be written with 2: 9.1×10^3.

Explain why he is wrong, and give the correct sum.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Ellis says: 9,145 has 4 significant digits and 4.8 has 2,

so the sum $9{,}145 + 4.8$ must be written with 2: 9.1×10^3.

Explain why he is wrong, and give the correct sum.

Adding uses decimal places, not a count of digits.

$9{,}145 + 4.8 = 9{,}149.8$

9,145 stops at the ones place, so round to ones: 9,150

Answer: $9{,}150 = 9.150 \times 10^3$

Independent Set

1. How many significant digits are in $6{,}030$ grams?
2. How many significant digits are in 0.00710 meters?
3. A rectangle measures 2.4×10^2 m by 3.15×10^2 m. Find its area, to the correct number of significant digits.
4. $(6.30 \times 10^4) + (2.8 \times 10^3)$ meters, both measured. Write the sum with the correct precision.
5. A car travels 1.80×10^2 km in 4.0 hours. Find its average speed, to the correct number of significant digits.
6. $(3.7 \times 10^{-2}) - (8.45 \times 10^{-4})$ grams, both measured. Write the difference with the correct precision.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due by the Unit 4 test.

Today's skills — all required

Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

MA.8.NSO.1.7

I can evaluate an expression that contains square roots, cube roots and integer exponents.

Warm-Up

1. How many significant digits are in 0.0250 ?
2. $(3 \times 10^{4})(2 \times 10^{-7})$
3. Find the value: $\sqrt{196}$
4. Evaluate: $20 - 3 \cdot 4$

Warm-Up

1. How many significant digits are in 0.0250 ? **3**
2. $(3 \times 10^{4})(2 \times 10^{-7})$ **6×10^{-3}**
3. Find the value: $\sqrt{196}$ **14**
4. Evaluate: $20 - 3 \cdot 4$ **8**

Vocabulary

radical

a root symbol together with everything under its bar: $\sqrt{36}$ is a square root, $\sqrt[3]{64}$ is a cube root

radicand

the number under the radical bar. The bar groups it, so whatever is under it is evaluated before the root is taken

principal square root

the nonnegative one. $\sqrt{49}$ is 7 — not −7, and not both, even though $(-7)^2$ is also 49

Notes

I. The radical bar is a grouping symbol

$$\sqrt{25} - \sqrt{36} + \sqrt{1} = 5 - 6 + 1 = 0$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4$$

$$\sqrt{9 + 16} = \sqrt{25} = 5, \quad \text{never} \quad 3 + 4$$

Everything under the bar is evaluated before the root is taken.

Notes

II. Square roots and cube roots are not the same shape

$$\sqrt{49} = 7 \qquad \sqrt[3]{64} = 4 \qquad \sqrt[3]{-64} = -4$$

$$\sqrt{\frac{1}{9}} = \frac{1}{3} \qquad \sqrt[3]{\frac{1}{125}} = \frac{1}{5}$$

An odd root keeps the sign of its radicand. A square root sign never gives a negative value.

Notes

III. Radicals alongside exponents

$$5^{-3} = \frac{1}{5^3} = \frac{1}{125} \qquad \sqrt{4^3 - 39} = \sqrt{25} = 5$$

$$\sqrt{4^3 - 39} \cdot 5^{-3} = 5 \cdot \frac{1}{125} = \frac{1}{25}$$

Finish each piece — the radical, the power — and only then put them together.

Example 1

$$\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$$

Evaluate.

Example 1

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4$$

$$\frac{1}{4} + 4 = \frac{17}{4}$$

Answer: 17/4

Your Turn

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

Evaluate.

Your Turn

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

Evaluate.

$$\left(-\frac{1}{3}\right)^2 = \frac{1}{9}$$

$$\sqrt[3]{2^2 + 4} = \sqrt[3]{8} = 2$$

$$\frac{1}{9} - 2 = -\frac{17}{9}$$

Answer: –17/9

Example 2

$$\sqrt[3]{27} - 1.4\left(\sqrt{3^2 - 5}\right)$$

Evaluate.

Example 2

$\sqrt[3]{27} = 3, \qquad 3^2 - 5 = 4$

$\sqrt{4} = 2, \qquad 1.4(2) = 2.8$

$3 - 2.8 = 0.2$

Answer: 0.2

Your Turn

$$\sqrt[3]{-8} + 2.5\left(\sqrt{5^2 - 9}\right)$$

Evaluate.

Your Turn

$$\sqrt[3]{-8} + 2.5\left(\sqrt{5^2 - 9}\right)$$

Evaluate.

$\sqrt[3]{-8} = -2$

$\sqrt{5^2 - 9} = \sqrt{16} = 4$

$-2 + 2.5 \times 4 = -2 + 10 = 8$

Answer: 8

Whiteboards · Question 1 of 9

$$\sqrt{121}$$

Answer it.

Whiteboards · Question 1 of 9

$$\sqrt{121}$$

$11 \times 11 = 121$, so $\sqrt{121} = 11$

Answer: 11

Whiteboards · Question 2 of 9

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

Answer it.

Whiteboards · Question 2 of 9

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

$$\text{Side} \times \text{side} = 169$$

$$13 \times 13 = 169, \quad \text{so} \quad \sqrt{169} = 13$$

Answer: 13 feet

Whiteboards · Question 3 of 9

$$\sqrt[3]{-125}$$

Answer it.

Whiteboards · Question 3 of 9

$$\sqrt[3]{-125}$$

$$(-5)\times(-5)\times(-5) = -125, \quad \text{so} \quad \sqrt[3]{-125} = -5$$

Answer: −5

Whiteboards · Question 4 of 9

$$\sqrt{2^3 + 17}$$

Answer it.

Whiteboards · Question 4 of 9

$$\sqrt{2^3 + 17}$$

Under the bar first: $2^3 + 17 = 8 + 17 = 25$

$\sqrt{25} = 5$

Answer: 5

Whiteboards · Question 5 of 9

$$\sqrt{25 + 144}$$

Which is the value?

A. 17

B. 169

C. 84.5

D. 13

Answer it.

Whiteboards · Question 5 of 9

$$\sqrt{25 + 144}$$

Which is the value?

Under the bar first: $25 + 144 = 169$

$\sqrt{169} = 13$

Answer: D — 13

Whiteboards · Question 6 of 9

$$\sqrt[3]{-\frac{1}{64}}$$

Which is the value?

A. −1/4

B. not a real number

C. −1/8

D. 1/4

Answer it.

Whiteboards · Question 6 of 9

$$\sqrt[3]{-\frac{1}{64}}$$

Which is the value?

$$\left(-\frac{1}{4}\right) \times \left(-\frac{1}{4}\right) \times \left(-\frac{1}{4}\right) = -\frac{1}{64}$$

so $\sqrt[3]{-\frac{1}{64}} = -\frac{1}{4}$

Answer: A — –1/4

Whiteboards · Question 7 of 9

$$\sqrt{\frac{4}{9}} + \sqrt[3]{\frac{8}{27}}$$

Answer it.

Whiteboards · Question 7 of 9

$$\sqrt{\frac{4}{9}} + \sqrt[3]{\frac{8}{27}}$$

$\sqrt{\frac{4}{9}} = \frac{2}{3}$ and $\sqrt[3]{\frac{8}{27}} = \frac{2}{3}$

$\frac{2}{3} + \frac{2}{3} = \frac{4}{3}$

Answer: 4/3

Whiteboards · Question 8 of 9

$$\sqrt{5^3 - 25} \cdot 10^{-3}$$

Answer it.

Whiteboards · Question 8 of 9

$$\sqrt{5^3 - 25} \cdot 10^{-3}$$

$$\sqrt{5^3 - 25} = \sqrt{125 - 25} = \sqrt{100} = 10$$

$$10^{-3} = \frac{1}{1{,}000}$$

$$10 \times \frac{1}{1{,}000} = \frac{10}{1{,}000} = \frac{1}{100}$$

Answer: 1/100

Whiteboards · Question 9 of 9

Simon says that $\sqrt[3]{-27}$ is not a real number,

because you cannot take the root of a negative.

Explain why he is wrong, and give the value.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Simon says that $\sqrt[3]{-27}$ is not a real number,

because you cannot take the root of a negative.

Explain why he is wrong, and give the value.

A cube root can be negative: three negatives multiply to a negative.

$(-3) \times (-3) \times (-3) = -27$, so $\sqrt[3]{-27} = -3$

Answer: -3

Independent Set

1. $\sqrt{225}$
2. $\sqrt[3]{-1}$
3. $\sqrt{6^2 + 13}$
4. $\sqrt[3]{\frac{1}{8}}$
5. $\sqrt{144} \cdot 2^{-3}$
6. $\left(-\frac{1}{2}\right)^3 + \sqrt{4^2 + 9}$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. This skill continues tomorrow: one assignment, due at the start of the class after 4.06.

Today's skills — all required

Evaluate radical expressions — D8J

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

MA.8.NSO.1.7

I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

Warm-Up

1. $\sqrt{5^2 - 9}$
2. How many significant digits are in 1.020 ?
3. Find the value: $\left(\frac{2}{3}\right)^{-2}$
4. Evaluate: $12 \div 3 \cdot 2$

Warm-Up

1. $\sqrt{5^2 - 9}$ **4**
2. How many significant digits are in 1.020 ? **4**
3. Find the value: $\left(\frac{2}{3}\right)^{-2}$ **9/4**
4. Evaluate: $12 \div 3 \cdot 2$ **8**

Vocabulary

grouping symbol

anything that holds part of an expression together and is finished first: parentheses, brackets, the radical bar and the fraction bar

nested

one grouping symbol inside another. The innermost one is finished first

multi-step expression

one that takes more than one operation to evaluate. Everything in this unit takes six steps or fewer

Notes

I. The structure, not a mnemonic

	Level	What it means
1	Grouping symbols	( ), [ ], the radical bar, the fraction bar
2	Powers and roots	the same level as each other
3	Multiply and divide	left to right
4	Add and subtract	left to right

Innermost grouping symbol first, then out.

Notes

II. Left to right means left to right

$$12 \div 3 \cdot 2 = 4 \cdot 2 = 8, \quad \text{never} \quad 12 \div 6 = 2$$

$$10 - 4 + 3 = 6 + 3 = 9, \quad \text{never} \quad 10 - 7 = 3$$

Multiplication and division are one level. Addition and subtraction are one level. Left to right, both times.

Notes

III. The fraction bar groups, top and bottom

$$\frac{2^3 - 5}{6^2 - 27} = \frac{3}{9} = \frac{1}{3}$$

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}} = \frac{1}{12} - \frac{1}{3} = -\frac{1}{4}$$

Finish the numerator and finish the denominator before you divide.

Example 1

$$4\left(8.63 - (3.2)^2\right) + \sqrt[3]{125}$$

Evaluate.

Example 1

$(3.2)^2 = 10.24$

$8.63 - 10.24 = -1.61, \qquad 4(-1.61) = -6.44$

$-6.44 + 5 = -1.44$

Answer: -1.44

Your Turn

$$3\left(5.4 - (2.1)^2\right) + \sqrt{49}$$

Evaluate.

Your Turn

$$3(5.4 - (2.1)^2) + \sqrt{49}$$

Evaluate.

Powers and roots: $(2.1)^2 = 4.41$ and $\sqrt{49} = 7$

Inside the parentheses: $5.4 - 4.41 = 0.99$

$3 \times 0.99 + 7 = 2.97 + 7 = 9.97$

Answer: 9.97

Example 2

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}}$$

Evaluate.

Example 2

$$\frac{2^3 - 5}{6^2} = \frac{3}{36} = \frac{1}{12}$$

$$\frac{4}{9} - \frac{1}{3} = \frac{1}{9}, \qquad \sqrt{\frac{1}{9}} = \frac{1}{3}$$

$$\frac{1}{12} - \frac{1}{3} = -\frac{1}{4}$$

Answer: –1/4

Your Turn

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

Evaluate.

Your Turn

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

Evaluate.

$$\frac{3^2 - 1}{4^2} = \frac{8}{16} = \frac{1}{2}$$

$$\sqrt{\frac{7}{16} - \frac{3}{16}} = \sqrt{\frac{4}{16}} = \frac{2}{4} = \frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2} = 1$$

Answer: 1

Whiteboards · Question 1 of 9

$$90 \div 5 \cdot 3$$

Answer it.

Whiteboards · Question 1 of 9

$$90 \div 5 \cdot 3$$

Left to right: $90 \div 5 = 18$

$18 \cdot 3 = 54$

Answer: 54

Whiteboards · Question 2 of 9

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

Answer it.

Whiteboards · Question 2 of 9

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

$$\sqrt{196} = 14$$

Perimeter: $4 \times 14 = 56$

Answer: 56 centimeters

Whiteboards · Question 3 of 9

$$16 - 5 + 4$$

Answer it.

Whiteboards · Question 3 of 9

$$16 - 5 + 4$$

Left to right: $16 - 5 = 11$

$11 + 4 = 15$

Answer: 15

Whiteboards · Question 4 of 9

$$\frac{3^3 - 21}{5^2 - 15}$$

Answer it.

Whiteboards · Question 4 of 9

$$\frac{3^3 - 21}{5^2 - 15}$$

Top: $3^3 - 21 = 27 - 21 = 6$

Bottom: $5^2 - 15 = 25 - 15 = 10$

$\frac{6}{10} = \frac{3}{5}$

Answer: 3/5

Whiteboards · Question 5 of 9

$$2 + 3 \cdot 4^2$$

Which is the value?

A. 146

B. 50

C. 26

D. 80

Answer it.

Whiteboards · Question 5 of 9

$$2 + 3 \cdot 4^2$$

Which is the value?

Power first: $4^2 = 16$

Then multiply: $3 \cdot 16 = 48$

Then add: $2 + 48 = 50$

Answer: B — 50

Whiteboards · Question 6 of 9

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

A. 4

B. 35.5

C. 0.5

D. 256

Answer it.

Whiteboards · Question 6 of 9

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

Top: $6^2 - 4 = 36 - 4 = 32$

Bottom: $2^3 = 8$

$32 \div 8 = 4$

Answer: A — 4

Whiteboards · Question 7 of 9

$$-2^2 + (-2)^2$$

Answer it.

Whiteboards · Question 7 of 9

$$-2^2 + (-2)^2$$

$-2^2 = -4$ (the exponent is on the 2 only)

$(-2)^2 = 4$

$-4 + 4 = 0$

Answer: 0

Whiteboards · Question 8 of 9

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

Answer it.

Whiteboards · Question 8 of 9

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$ and $3^{-2} = \frac{1}{9}$

$\frac{1}{8} + \frac{1}{9} = \frac{9}{72} + \frac{8}{72} = \frac{17}{72}$

$12 \times \frac{17}{72} = \frac{17}{6}$

Answer: 17/6

Whiteboards · Question 9 of 9

Roscoe says that $\left(\sqrt{n} - 11\right)^2$ is always positive,

because a square is always positive.

Is he right? Give a value of n that settles it.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Roscoe says that $(\sqrt{n} - 11)^2$ is always positive,

because a square is always positive.

Is he right? Give a value of n that settles it.

When $n = 121$: $\sqrt{121} - 11 = 11 - 11 = 0$

$0^2 = 0$, and 0 is not positive.

Answer: No — at n = 121 the value is 0

Independent Set

1. $30 - 4 \cdot 5 + 2$
2. $36 \div 6 \cdot 2$
3. $3 + 5(6 - 2)^2$
4. $\dfrac{5^2 - 1}{2^2 + 2}$
5. $-4^2 + (-4)^2$
6. $\sqrt{100 - 19} + 2^{-2}$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due by the Unit 4 test.

Today's skills — all required

Evaluate radical expressions — D8J

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 7–8

Real-World Order of Operations

MA.8.NSO.1.7

I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

Warm-Up

1. $-5^2 + (-5)^2$
2. $\sqrt{144} + 2^{-1}$
3. Find the value: $\left(\frac{3}{4}\right)^{-1}$
4. Evaluate $2x^2$ when $x = 3$.

Warm-Up

1. $-5^2 + (-5)^2$ **0**
2. $\sqrt{144} + 2^{-1}$ **25/2**
3. Find the value: $\left(\frac{3}{4}\right)^{-1}$ **4/3**
4. Evaluate $2x^2$ when $x = 3$. **18**

Vocabulary

formula

a rule written with letters, such as $K = \frac{1}{2}mv^2$,

that becomes a number once each letter is replaced by a value

substitute

replace a letter with its value — in parentheses, so that a negative value keeps its sign and a power stays attached to the whole value

unit

what the answer is counted in: feet, dollars, joules. A real-world answer without a unit is not an answer

Notes

I. Substituting: the value goes in parentheses

$$K = \frac{1}{2}mv^2 \quad \text{with} \quad m = 60 \quad \text{and} \quad v = 10:$$

$$\frac{1}{2}(60)(10)^2 = \frac{1}{2}(60)(100) = 3{,}000 \quad \text{joules}$$

$$\text{with} \quad x = -4: \quad x^2 = (-4)^2 = 16, \quad \text{never} \quad -4^2 = -16$$

Substitute in parentheses, then evaluate by the structure — powers before products.

Notes

II. Key words do not decide the operation

15 dollars plus 4 dollars for each of 6 gigabytes:

$$15 + 4(6) = 39$$

15 dollars for each of 4 lines, plus 6 dollars of tax:

$$15(4) + 6 = 66$$

The same words, the same numbers, different structures. Read the situation.

Notes

III. Round once, at the end — and say the unit

$$6\sqrt{200} = 84.8528\ldots \approx 84.85 \quad \text{feet}$$

rounding the side to 14 first: $6(14) = 84$ feet

Carry the exact value, round once at the end, and write the unit.

Example 1

Kinetic energy is $K = \frac{1}{2}mv^2$,

for mass m in kilograms and speed v in meters per second.

Find K for a car of mass 1,200 kg travelling at 15 m/s.

Give the answer with its unit.

Example 1

$$K = \frac{1}{2}(1{,}200)(15)^2$$

$$= \frac{1}{2}(1{,}200)(225)$$

$$= 600(225) = 135{,}000$$

Answer: 135,000 joules

Your Turn

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

Your Turn

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

$$v^2 = 12^2 = 144$$

$$K = \frac{1}{2} \times 850 \times 144 = 425 \times 144 = 61{,}200$$

Answer: 61,200 joules

Example 2

A square floor of side 9 feet is carpeted.

Carpet is 2.89 dollars a square foot and padding is 0.75 dollars a square foot.

Installation is a flat 95 dollars. Find the total cost.

Example 2

$$9^2(2.89 + 0.75) + 95$$

$$= 81(3.64) + 95$$

$$= 294.84 + 95 = 389.84$$

Answer: 389.84 dollars

Your Turn

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

Your Turn

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

Area: $6^2 = 36$

Tile: $36 \times 4.25 = 153$

Total: $153 + 13.48 = 166.48$

Answer: 166.48 dollars

Whiteboards · Question 1 of 9

Evaluate $3x^2$ when $x = -2$.

Answer it.

Whiteboards · Question 1 of 9

Evaluate $3x^2$ when $x = -2$.

$$3x^2 = 3(-2)^2$$

$$(-2)^2 = 4$$

$$3 \times 4 = 12$$

Answer: 12

Whiteboards · Question 2 of 9

A cyclist has a mass of 70 kilograms and rides at 8 meters per second.

Her bicycle cost 480 dollars.

Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v.

Find her kinetic energy.

Answer it.

Whiteboards · Question 2 of 9

A cyclist has a mass of 70 kilograms and rides at 8 meters per second.

Her bicycle cost 480 dollars.

Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v.

Find her kinetic energy.

$$K = \frac{1}{2} \times 70 \times 8^2 = 35 \times 64 = 2{,}240$$

Answer: 2,240 joules

Whiteboards · Question 3 of 9

Renting a kayak costs 17 dollars plus 9 dollars for each hour.

Find the cost of 3 hours.

Answer it.

Whiteboards · Question 3 of 9

Renting a kayak costs 17 dollars plus 9 dollars for each hour.

Find the cost of 3 hours.

$$9 \times 3 = 27$$

$$17 + 27 = 44$$

Answer: 44 dollars

Whiteboards · Question 4 of 9

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

Answer it.

Whiteboards · Question 4 of 9

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

Area: $8^2 = 64$

Tile: $64 \times 3.50 = 224$

Total: $224 + 60 = 284$

Answer: 284 dollars

Whiteboards · Question 5 of 9

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

A. 180

B. 120

C. 360

D. 3,600

Answer it.

Whiteboards · Question 5 of 9

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

$v^2 = 3^2 = 9$

$\frac{1}{2} \times 40 \times 9 = 20 \times 9 = 180$

Answer: A — 180

Whiteboards · Question 6 of 9

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

A. $12(1.25) + 38$

B. $12^2 + 1.25 + 38$

C. $12^2(1.25 + 38)$

D. $12^2(1.25) + 38$

Answer it.

Whiteboards · Question 6 of 9

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

Area 12^2 square feet, at 1.25 dollars each: $12^2 \times 1.25$

The 38 dollars is added once: $12^2 \times 1.25 + 38 = 218$

Answer: D — $12^2(1.25) + 38$

Whiteboards · Question 7 of 9

A square garden has 4 sides, each $\sqrt{150}$ feet long.

Find its perimeter, to the nearest hundredth of a foot.

Answer it.

Whiteboards · Question 7 of 9

A square garden has 4 sides, each $\sqrt{150}$ feet long.

Find its perimeter, to the nearest hundredth of a foot.

$$\sqrt{150} \approx 12.2474$$

$$4 \times 12.2474 \approx 48.99$$

Answer: 48.99 feet

Whiteboards · Question 8 of 9

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

Answer it.

Whiteboards · Question 8 of 9

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

Top: $7^2 - 4 = 49 - 4 = 45$

$45 \div 5 = 9$

Answer: 9

Whiteboards · Question 9 of 9

One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings.

Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery.

Find both totals, and say why the key words could not have told you.

Write your answer in sentences.

Whiteboards · Question 9 of 9

One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings.

Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery.

Find both totals, and say why the key words could not have told you.

First order: $13 + 2 \times 5 = 13 + 10 = 23$

Second order: $13 \times 2 + 5 = 26 + 5 = 31$

Both say "plus" and "for each". What repeats is different.

Answer: 23 dollars and 31 dollars

Independent Set

1. $6x^2$ when $x = -2$
2. $\frac{1}{2}mv^2$ when $m = 90$ and $v = 4$
3. $\frac{a^2 - b}{c}$ when $a = 9,\ b = 6,\ c = 5$
4. A square room of side 10 feet is carpeted at 2.65 dollars a square foot, with a flat 55 dollars for fitting. Find the total cost.
5. $2(x + y)^2 - 3$ when $x = 4$ and $y = 1$
6. A gym charges 25 dollars to join plus 8 dollars for each of 6 classes. Find the total cost.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due by the unit test.

Today's skills — all required

Evaluate variable expressions: word problems — DPG